

Chapter 1: Sequences and Limits

1.1 Convergence

Body text discussing convergence of sequences.

Definition 1.1.1 (Cauchy Sequence). A sequence is Cauchy if...

for all $\epsilon > 0$ there exists N such that $|a_n - a_m| < \epsilon$

1.2 Continuity

Theorem 1.2.1. If f is continuous on $[a,b]$, then f is bounded.

Proof. Suppose not...

Example 1.2.1. Consider $f(x) = x^2$ on $[0,1]$.

Chapter 2: Computer Science Fundamentals

2.1 Algorithms and Complexity

```
def binary_search(arr, target):  
    lo, hi = 0, len(arr) - 1  
    while lo <= hi:  
        mid = (lo + hi) // 2
```

The time complexity is $O(\log n)$.

2.1.1 Big-O Notation

We say $f(n) = O(g(n))$ if there exist constants c, n_0 such that...

Exercise 2.1. Prove that $3n^2 + 5n = O(n^2)$.

Appendix A: Notation Reference

A.1 Set Notation

This appendix lists common notation used throughout.